# INTAN DWI ANGGREINI

Applied Data Science Student | Data Engineering | Machine Learning | MLOps

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## SUMMARY

Applied Data Science student (5th semester) at Politeknik Elektronika Negeri Surabaya (PENS) who builds end-to-end data and machine learning systems: an MLOps credit risk platform (FastAPI, MLflow, DVC, Docker), a star-schema data warehouse with UMAP-HDBSCAN risk segmentation, fault-tolerant Apache Airflow pipelines and a Spark, HDFS and Hive big data workflow. Communicates insights through Power BI, Tableau, Grafana and D3.js dashboards, with a networking and Linux administration foundation (MTCNA certified).

## EDUCATION

**Electronic Engineering Polytechnic Institute of Surabaya (PENS)** — 2024 - Present
*Bachelor of Applied Data Science (D4 Sains Data Terapan); GPA 3.57/4.00* — Surabaya, Indonesia

**SMK Negeri 1 Kediri (Kediri Vocational High School 1)** — 2021 - 2024
*Computer Network and Telecommunication Engineering* — Kediri, Indonesia

- Average report grade 89/100; ranked among the top 5 students in the major throughout the three-year program.

## TECHNICAL SKILLS

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|---|---|
| **Programming** | Python, SQL, R, JavaScript, PHP, Rust, Kotlin |
| **Machine Learning** | scikit-learn, Spark MLlib, Deep Learning, Computer Vision, K-Means, UMAP, HDBSCAN |
| **MLOps and DevOps** | MLflow, DVC, Docker, Docker Compose, Git |
| **Data Engineering** | Apache Airflow, Apache Spark (PySpark), HDFS, Apache Hive, Apache Kafka, Debezium, Pentaho Data Integration, ETL, Data Warehousing (Star Schema) |
| **Databases** | PostgreSQL, MySQL, MongoDB |
| **Web and APIs** | FastAPI, Flask, REST API, React, PHP, Bootstrap, Leaflet.js |
| **Data Visualization** | Power BI, Tableau, Grafana, Looker Studio (Google Data Studio), D3.js, Kibana |
| **Systems and Network** | Linux (Debian, Ubuntu, Red Hat), MikroTik, Cisco, wired and wireless network installation |
| **Soft Skills** | Problem Solving, Critical Thinking, Communication, Teamwork, Leadership |

## PROJECTS

**RiskFinder - Machine Learning Credit Risk Analysis Platform** — Team Lead, Full-Stack and ML Engineer
**Tech:** Python, FastAPI, React, PostgreSQL, MLflow, DVC, Docker
github.com/intananggreini99/riskfinder-project

- Led a team building a credit risk system that predicts borrower default to help banks, P2P lending and fintech platforms make faster, more objective lending decisions.
- Built the training pipeline (KNN imputation, outlier capping, feature selection, imbalance handling, model comparison, hyperparameter tuning), tracking experiments in MLflow and versioning data and artifacts with DVC.
- Developed a FastAPI and React app with JWT role-based access for training, model registry and activation, risk prediction and monitoring (ROC-AUC, confusion matrix, prediction history), deployed with Docker and PostgreSQL.

**Credit Risk Segmentation and Anomaly Detection with UMAP-HDBSCAN on a Data Warehouse** — Personal Project
**Tech:** Python, scikit-learn, UMAP, HDBSCAN, PostgreSQL, Pentaho Data Integration, PHP, D3.js
github.com/intananggreini99/UMAP-HDBSCAN_RiskSegmentation_and_CreditAnomalyDetection

- Built an ETL pipeline (Pentaho Data Integration and SQL) loading 1,000 loan applications from an operational data store into a PostgreSQL star schema (8 dimension and 2 fact tables).
- Combined UMAP (10 financial features) with HDBSCAN and benchmarked it against 7 other algorithms on 7 prioritized metrics; it ranked first (Silhouette 0.781, Davies-Bouldin 0.34) and isolated 37.9% of applications as anomalies.
- Derived 7 tiered risk segments, including a "Hidden Prime" segment fit for automatic approval, shown in an interactive PHP and D3.js dashboard (25% default rate, 43% average ROI, IDR 83.78 billion total profit).

**Airflow Weather Data Pipeline with Observability and Alerting** — Team Project

**Tech:** Apache Airflow 3, Python, PostgreSQL, Redis, Celery, Docker Compose, Open-Meteo API

github.com/mudrik696/airflow-project

- Co-developed a weather data pipeline for Surabaya on Apache Airflow with 9 DAGs covering extraction, transformation, branching data-quality checks and cross-DAG orchestration.
- Four layers of failure handling: automatic retries, centralized logging to PostgreSQL, SMTP email alerts and a custom UI to investigate and re-run failed tasks; runs on Docker Compose (CeleryExecutor, Redis).

**Apple Leaf Disease Classification on a Big Data Stack** — Big Data Project

**Tech:** PySpark, Spark MLlib, scikit-learn, scikit-image, HDFS, Apache Hive, PostgreSQL, Docker Compose

github.com/intananggreini99/AppleDiseaseClassification

- Set up a 9-container Docker environment (Spark standalone cluster, HDFS, Hive metastore and HiveServer2, PostgreSQL); ingested leaf images into HDFS and extracted features in parallel with Spark UDFs and scikit-image into Parquet, Hive and PostgreSQL tables.
- Benchmarked Spark MLlib against scikit-learn on identical features, comparing training time and evaluation metrics, with per-stage durations logged to PostgreSQL.

**COVID-19 Distribution Dashboard for Indonesia** — Web Dashboard

**Tech:** Python, K-Means, PHP, MySQL, D3.js, Leaflet, Grafana

github.com/intananggreini99/Dasboard_Covid19_PHP-D3JS

- Built an interactive dashboard of COVID-19 cases across 34 provinces (3,493 records, March to June 2020) with PHP and MySQL JSON endpoints, D3.js charts and a Leaflet choropleth map; grouped provinces into 4 risk levels with K-Means and recreated the view in Grafana.

**School Management Information System, TKJ Department of SMKN 1 Kediri** — Personal Project

**Tech:** PHP 8, MySQL 8, Bootstrap 5, JavaScript, Docker Compose

github.com/intananggreini99/MIS_WebDinamis_TKJSMEKSA

- Developed a web MIS with session login and role-based access (admin, teacher, student) to manage users, students, teachers, subjects and the school agenda, including photo uploads; containerized Apache, PHP, MySQL and phpMyAdmin with Docker Compose.

**Business Intelligence Dashboards** — Dashboards

**Tech:** Power BI, Tableau, Grafana, Looker Studio

**Power BI:** Phi Minimart report | **Looker Studio:** Toko Kita report | **Grafana:** github.com/intananggreini99/grafanaDashboard
**Tableau:** public.tableau.com/app/profile/intan.dwi.anggreini

- Designed sales and regional dashboards: Phi Minimart sales performance (Power BI and Grafana), Toko Kita sales and profit report (Looker Studio), and the distribution of high schools, universities and MSMEs in Kediri City (Tableau Public).

## EXPERIENCE

**Staff, Human Capital Division - IEEE Student Branch PENS** — 2026 - Present
*Politeknik Elektronika Negeri Surabaya* — Surabaya, Indonesia

- Support member development and welfare programs for the student branch, and presented the division's role and programs to members.

**Guide (Pemandu) - LKMM PENS 2025, Student Management Skills Training** — 2025 - Present
*Politeknik Elektronika Negeri Surabaya* — Surabaya, Indonesia

- Facilitate management and leadership training sessions for first-year students: deliver material, lead class sessions and collect participant feedback.

**Staff, Computer Center (Puskom) and Personnel (Kepegawaian) divisions - Politeknik Negeri Malang, PSDKU Kediri** — Jun 2023 - Nov 2023
*Vocational internship (PKL)* — Kediri, Indonesia

- Completed a six-month vocational internship supporting the campus's day-to-day administrative operations.

## CERTIFICATION

**MikroTik Certified Network Associate (MTCNA)**, MikroTik — 2024

## LANGUAGES

Indonesian, English